

SMART DISASTER RELIEF INVENTORY, ALLOCATION AND DECISION SUPPORT SYSTEM USING C

CSA0203 – C PROGRAMMING | BACHELOR OF ENGINEERING

Submitted by: A.GUNASEKAR(192525302),P.JEEVAN KUMAR(192572090),C.SUSHANTH(192412418);

Under the Supervision of Dr. S. K. Saravanan

SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai – 602 105

PROJECT SUMMARY

Natural disasters such as floods, cyclones, earthquakes, landslides and tsunamis create an urgent need for food, drinking water, medicines, blankets, rescue equipment, temporary shelters and other essential resources. Manual inventory management can result in inaccurate records, duplicate entries, stock shortages, delayed distribution and loss of important information. The Smart Disaster Relief Inventory and Distribution Management System using C was developed as a menu-driven application to provide a structured and efficient solution for monitoring and managing relief resources. The system maintains Resource ID, Resource Name, Category, Quantity Available, Minimum Required Quantity, Distribution Centre, Priority Level and Distribution Demand. It supports adding, updating, deleting, searching, sorting, merging, analysing, distributing and permanently storing resource information. Resources can be searched using identifiers or categories and sorted according to quantity, priority and distribution centre. Records from multiple relief centres can be merged while preventing duplicate Resource IDs. The analysis module compares stock with required quantities and identifies resources as Adequate, Low or Critical, helping administrators recognize shortages and immediate replenishment requirements. The distribution module checks available stock before allocation and prevents distribution beyond the quantity available. File handling enables inventory data to be saved and loaded and allows disaster-resource reports to be generated. The project demonstrates practical use of operators, decision-making statements, switch-case, loops, arrays, strings, structures, functions, pointers, recursion, searching, sorting, merging and file handling in C. Testing covered valid resource addition, duplicate IDs, searching, shortage analysis, distribution, excess distribution, sorting, file saving and report generation, with the documented test cases producing PASS results. Overall, the project applies core C programming concepts to a practical disaster-management problem and provides a foundation for future enhancement using databases, cloud storage, GPS tracking, IoT sensors, mobile applications and artificial intelligence.

KEY FUNCTIONALITIES

Area	Functionality	Purpose
Resource Management	Add, update, delete, display	Maintain accurate inventory records
Search & Sorting	ID/category search; quantity, priority & centre sorting	Quick resource identification and organization
Analysis	Adequate / Low / Critical classification	Detect shortages and replenishment needs
Distribution	Stock validation and allocation	Prevent over-allocation and manage stock outflow
Multi-Centre Merge	Merge records with duplicate-ID prevention	Consolidate relief-centre inventory
File & Reports	Save, load and generate reports	Provide persistence and summarized information

TECHNOLOGY AND C CONCEPTS

Environment: C programming, GCC compiler, Windows/Linux, CodeBlocks/Dev-C++/VS Code and local file storage. **Concepts:** operators, decision-making, switch-case, loops, arrays, strings, structures, functions, pointers, recursion, searching, sorting, merging and file handling.

OUTCOME

The completed system provides a simple and modular approach to disaster-relief inventory control. It improves visibility of available resources, supports timely distribution and highlights critical shortages, while demonstrating how fundamental and advanced C programming techniques can be integrated into a real-world application.